PROJECT SUMMARY · HEALTH DATA ANALYTICS

Diabetes Risk Factors: Inference and Prediction on BRFSS Data

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229,781	17.3%	0.80	18% → 49%
survey respondents (BRFSS 2015)	prevalence of (pre)diabetes	AUC of the logistic model	recall after threshold tuning

Question

Which clinical, behavioural and socioeconomic characteristics are associated with diabetes, and how well can a simple, interpretable model identify people at risk? The project combines exploratory analysis, hypothesis testing and a logistic regression classifier, with particular attention to class imbalance, which is the central practical issue in medical screening.

Data and preparation

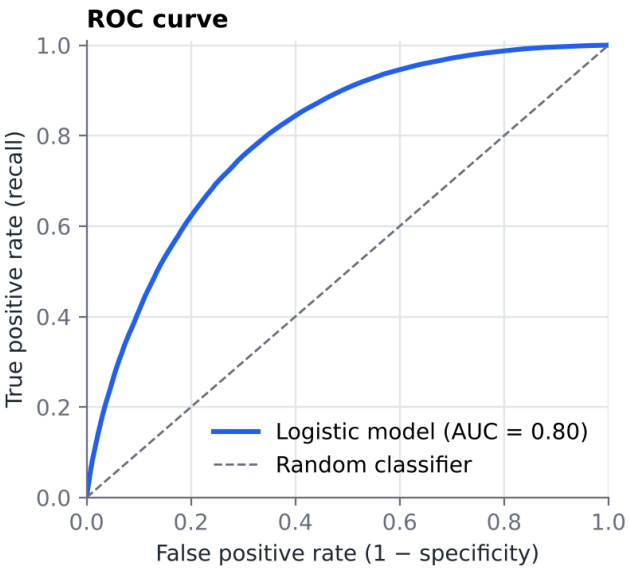
The data come from the Behavioral Risk Factor Surveillance System (BRFSS), the annual CDC health survey, in the public *Diabetes Health Indicators* version on Kaggle. Cleaning was done in Python (pandas): duplicate records were removed (253,680 → 229,781 rows), variables were renamed consistently, and the three-level outcome (no diabetes / prediabetes / diabetes) was recoded as binary, with prediabetes and diabetes forming the positive class. No values were missing. All statistical analysis and modelling were carried out in R.

What the data show

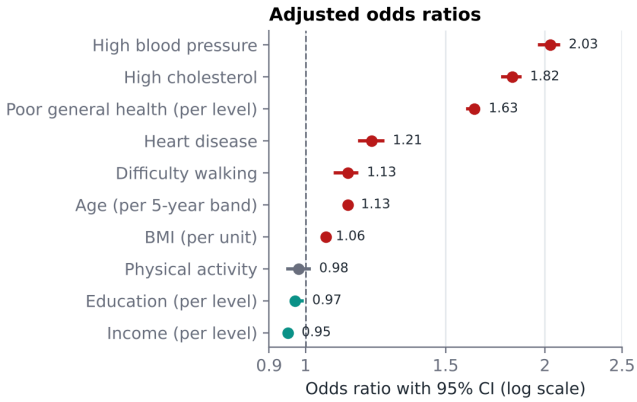
- Diabetic respondents have a higher mean BMI (31.8 vs 28.0) and belong to older age bands; both differences are highly significant (Welch t-tests, $p < 2.2e\text{-}16$).
- Prevalence is 28.1% among people with high blood pressure versus 8.3% without it, and 26.0% versus 10.4% for high cholesterol (χ^2 tests, $p < 2.2e\text{-}16$).
- The strongest correlates are self-rated general health ($r = 0.28$), high blood pressure (0.26), BMI (0.21) and difficulty walking (0.21); income (−0.15), education (−0.11) and physical activity (−0.10) are inversely related. All VIFs are below 2, so multicollinearity is not a concern.

Model

A logistic regression with 18 predictors (McFadden pseudo-$R^2 = 0.185$) reaches an AUC of 0.80. Holding the other variables fixed, high blood pressure roughly doubles the odds of diabetes (OR 2.03), high cholesterol raises them by 82% and each step of worse self-rated health by 63%; higher income and education are mildly protective.



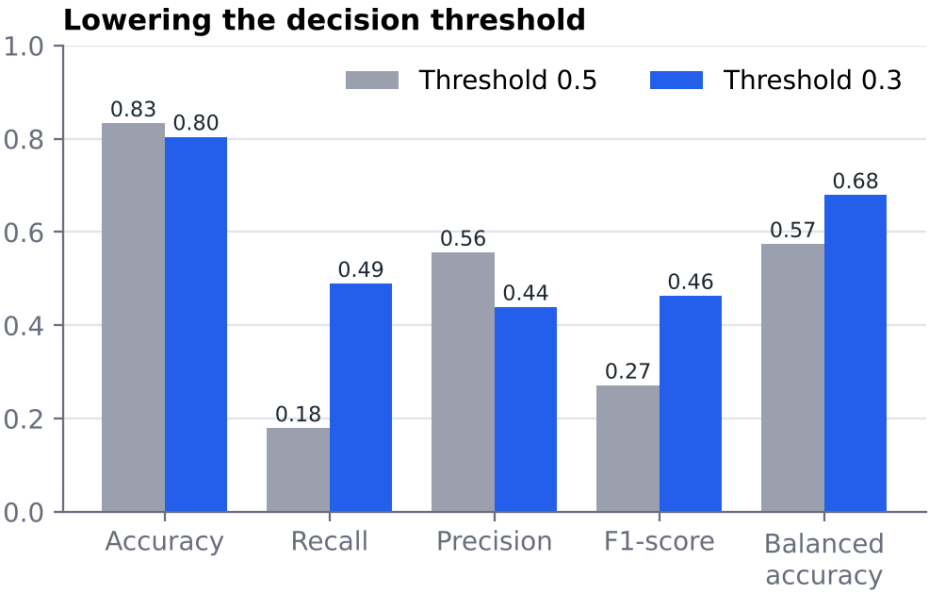
ROC curve of the full model.



Adjusted odds ratios. Red: higher risk; green: lower risk; grey: not significant.

Dealing with class imbalance

With the default 0.5 cut-off the model looks accurate (83.3%) but finds only 17.9% of diabetic respondents: it mostly predicts the majority class. Because a missed case is far more costly than a false alarm in screening, the threshold was lowered to 0.3. Accuracy falls slightly to 80.4%, but recall rises to 48.9% and F1 from 0.27 to 0.46. Results hold out of sample: on a 70/30 train–test split the model keeps an AUC of about 0.80 (sensitivity 0.49, specificity 0.87 at 0.3), and 10-fold cross-validation confirms its stability. Rebalancing with oversampling, undersampling, SMOTE and class weights leaves the AUC at 0.80–0.81, showing that imbalance affects the choice of threshold more than the model's ability to rank individuals.



Classification metrics at the default and the tuned threshold.

Interpretation and limits

- The data are observational and self-reported: the associations are not causal and may carry reporting bias. The apparently protective effect of heavy alcohol use, for instance, is best explained by confounding (heavy drinkers in the sample are younger and leaner) rather than by a real benefit.

- Age, health status, income and education are ordinal codes, so their coefficients refer to moving up one category.
- With n > 229,000 almost every effect is significant, so conclusions rely on odds ratios and confidence intervals rather than p-values. The calibration plot and Hosmer–Lemeshow test indicate reasonably reliable predicted probabilities.
- The model is a statistical screening aid, not a diagnostic tool. Natural next steps are non-linear models (random forest, gradient boosting) and richer clinical data.

> English summary of the full report, which is written in Italian (diabete.pdf). Charts in this summary were redrawn with English labels from the figures reported in the analysis.